

# Mathematical Guide

All Models, Losses, Strategies, and Uncertainty Methods  
in the AutoML Package

AutoML Research Project

April 2026 (rev. post Phase 8)

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# Part I

## Framework

### 1 Introduction and Notation

This document provides a complete mathematical specification of every model, loss function, selection strategy, uncertainty method, and evaluation metric in the AutoML package. All sections describe currently implemented behavior unless explicitly flagged otherwise (e.g., Laplace-approximation BNN in §24).

#### 1.1 Notation

| Symbol                        | Meaning                                           |
|-------------------------------|---------------------------------------------------|
| $\mathbf{x} \in \mathbb{R}^d$ | Input feature vector ( $d$ features)              |
| $y \in \mathbb{R}$            | Scalar regression target                          |
| $N$                           | Number of training samples                        |
| $K$                           | Number of classes / bins / mixture components     |
| $D$                           | Number of hidden layers (depth) in FlexibleNN     |
| $M$                           | Number of ensemble members or MC samples          |
| $\pi_k(\mathbf{x})$           | Probability of class $k$ given input $\mathbf{x}$ |
| $\mu_k(\mathbf{x})$           | Mean prediction of regression head $k$            |
| $\sigma_k^2(\mathbf{x})$      | Variance prediction of regression head $k$        |
| $\hat{y}(\mathbf{x})$         | Final point prediction                            |
| $\hat{\sigma}^2(\mathbf{x})$  | Final variance prediction                         |
| $\boldsymbol{\theta}$         | All learnable parameters                          |
| $\boldsymbol{\phi}$           | Parameters of a selection / gating subnetwork     |
| $\tau$                        | Temperature parameter (Gumbel-Softmax)            |
| $\alpha$                      | Miscoverage level for prediction intervals        |
| $\beta$                       | Variance weighting exponent ( $\beta$ -NLL)       |
| $k(\mathbf{x}, \mathbf{x}')$  | Kernel / covariance function (GP)                 |
| $\eta$                        | Learning rate                                     |

Tensor shapes are written in parentheses:  $(B, K)$  means batch size  $B$ ,  $K$  columns. We write  $[n]$  for  $\{1, 2, \dots, n\}$ .

### 2 Uncertainty Quantification: A Unified View

Every model in this package produces (or can produce) an uncertainty estimate  $\hat{\sigma}^2(\mathbf{x})$ . This section provides the taxonomy *before* describing individual models, so the reader has the mental model up front.

## 2.1 Taxonomy of Uncertainty Sources

| Method                 | How $\hat{\sigma}^2$ is produced               | What it captures           | Input-dep?    |
|------------------------|------------------------------------------------|----------------------------|---------------|
| Learned variance (NLL) | Second output of NN, trained via NLL           | Aleatoric                  | Yes           |
| Law of Total Variance  | Aggregation of per-class $(\mu_k, \sigma_k^2)$ | Aleatoric + epistemic-like | Yes           |
| MC Dropout             | Empirical variance of $M$ stochastic passes    | Epistemic (approx.)        | Yes           |
| Deep Ensembles         | Variance across $M$ independent models         | Aleatoric + epistemic      | Yes           |
| Binned Residual Std    | Variance of training residuals in bins         | Aleatoric (discretized)    | Partially     |
| Lookup Mapper          | Per-probability-bin variance                   | Aleatoric (discretized)    | Partially     |
| GP Posterior           | Analytical posterior variance                  | Aleatoric + epistemic      | Yes           |
| Conformal              | Calibration-set residual quantile              | Coverage guarantee         | No (marginal) |
| Constant               | Global scalar, not learned                     | None (placeholder)         | No            |

## 2.2 Learned Variance via NLL

The network outputs  $(\mu(\mathbf{x}), \log \sigma^2(\mathbf{x}))$  and is trained with the Gaussian NLL (§11.1). The gradient w.r.t.  $\log \sigma^2$ :

$$\frac{\partial \mathcal{L}}{\partial \log \sigma^2} = \frac{1}{2} \left( 1 - \frac{(y - \mu)^2}{\sigma^2} \right),$$

drives  $\sigma^2$  toward  $\mathbb{E}[(y - \mu)^2 \mid \mathbf{x}]$ , the true conditional variance, at convergence.

**Used by:** ProbReg (per-head), Plain NN (**PROBABILISTIC**), NN Mapper, Deep Ensembles (per member), NGBoost, tree models (custom objective).

## 2.3 Law of Total Variance

Given per-class predictions  $\{(\mu_k, \sigma_k^2)\}_{k=1}^K$  weighted by  $\{\pi_k\}_{k=1}^K$ :

$$\hat{\sigma}^2 = \underbrace{\sum_k \pi_k \sigma_k^2}_{\text{expected within-class variance}} + \underbrace{\sum_k \pi_k (\mu_k - \hat{y})^2}_{\text{between-class variance}}. \quad (1)$$

The first term is aleatoric (noise within each class). The second term captures disagreement among class means. When all  $\mu_k$  agree, the second term vanishes and we get pure aleatoric uncertainty.

**Used by:** ProbReg (central novelty), MDN, Deep Ensembles (with uniform  $\pi_m = 1/M$ ).

## 2.4 MC Dropout

Gal & Ghahramani (2016): keep dropout active at test time and run  $M$  forward passes:

$$\hat{y}_m = f_{\theta, \mathbf{z}_m}(\mathbf{x}), \quad \mathbf{z}_m \sim \text{Bernoulli}(1 - p)^L, \quad m = 1, \dots, M, \quad (2)$$

$$\hat{\mu} = \frac{1}{M} \sum_{m=1}^M \hat{y}_m, \quad \hat{\sigma}^2 = \frac{1}{M} \sum_{m=1}^M (\hat{y}_m - \hat{\mu})^2. \quad (3)$$

**Used by:** Plain NN (with MC\_DROPOUT).

## 2.5 Binned Residual Standard Deviation

A post-hoc method. After training:

1. Compute residuals:  $r_i = y_i - \hat{y}_i$  for all training samples.
2. Bin predictions into  $B$  bins by  $\hat{y}_i$ .
3. For each bin  $b$ :  $\hat{\sigma}_b = \text{std}(\{r_i : \hat{y}_i \in b\})$ .
4. At test time:  $\hat{\sigma}(\mathbf{x}) = \hat{\sigma}_{b(\hat{y}(\mathbf{x}))}$ .

**Partially input-dependent:** piecewise constant (one value per bin).

**Used by:** Linear Regression, XGBoost, LightGBM, CatBoost, Plain NN (with BINNED\_RESIDUAL\_STD).

## 2.6 Summary: How Each Model Produces Uncertainty

| Model                | Primary UQ Method                           | Alternatives                            |
|----------------------|---------------------------------------------|-----------------------------------------|
| Linear Regression    | Binned Residual Std                         | —                                       |
| Plain NN             | Depends on <code>uncertainty_method</code>  | CONSTANT / PROBABILISTIC / MC_DROPOUT / |
| XGBoost              | Custom NLL objective (learned $\sigma^2$ )  | Binned Residual Std                     |
| LightGBM             | Custom NLL objective (learned $\sigma^2$ )  | Binned Residual Std                     |
| CatBoost             | Custom NLL objective (learned $\sigma^2$ )  | Binned Residual Std                     |
| ProbReg              | Law of Total Variance                       | + Conformal wrapper                     |
| ClassReg (Lookup)    | Per-bin variance from lookup table          | Conformal wrapper                       |
| ClassReg (Spline)    | Constant residual variance                  | —                                       |
| ClassReg (NN Mapper) | Learned $\sigma^2$ via NLL                  | —                                       |
| FlexibleNN           | Same as Plain NN                            | —                                       |
| GP                   | Posterior variance (analytical)             | —                                       |
| Deep Ensembles       | Ensemble disagreement + mean aleatoric      | —                                       |
| NGBoost              | Natural gradient learned $\sigma^2$         | —                                       |
| MDN                  | Mixture variance (law of total variance)    | —                                       |
| Quantile Regression  | Quantile-based intervals                    | —                                       |
| CQR                  | Quantile regression + conformal calibration | —                                       |

## Part II

### Currently Supported Models

#### 3 Linear Regression

Ordinary least squares:

$$\hat{y} = \mathbf{w}^\top \mathbf{x} + b, \quad (\mathbf{w}, b) = \arg \min_{\mathbf{w}, b} \sum_{i=1}^N (y_i - \mathbf{w}^\top \mathbf{x}_i - b)^2.$$

**Uncertainty:** No native uncertainty. Uses post-hoc BINNED\_RESIDUAL\_STD (§2.5).

#### 4 Gradient-Boosted Trees (XGBoost, LightGBM, CatBoost)

All three share the same additive ensemble structure:

$$\hat{y}(\mathbf{x}) = \sum_{t=1}^T \eta \cdot f_t(\mathbf{x}), \quad f_t \in \mathcal{F},$$

where  $\mathcal{F}$  is the space of decision trees,  $\eta$  is the learning rate (shrinkage), and each tree  $f_t$  is fit to the pseudo-residuals.

##### 4.1 Regularized Objective

At step  $t$ :

$$\mathcal{L}^{(t)} = \sum_{i=1}^N \left[ g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t(\mathbf{x}_i)^2 \right] + \gamma T_{\text{leaves}} + \frac{1}{2} \lambda \sum_{j=1}^{T_{\text{leaves}}} w_j^2, \quad (4)$$

where  $g_i = \partial \mathcal{L} / \partial \hat{y}_i$  and  $h_i = \partial^2 \mathcal{L} / \partial \hat{y}_i^2$ .

##### 4.2 Custom NLL Objective for Probabilistic Output

For trees predicting  $(\mu, \log \sigma^2)$ , custom gradients and Hessians:

**Mean output:**

$$g_\mu = \frac{\mu - y}{\sigma^2}, \quad h_\mu = \frac{1}{\sigma^2}. \quad (5)$$

**Log-variance output:**

$$g_{\log \sigma^2} = \frac{1}{2} \left( 1 - \frac{(y - \mu)^2}{\sigma^2} \right), \quad h_{\log \sigma^2} = \frac{1}{2} \quad (\text{Fisher information}). \quad (6)$$

The Fisher information  $h = \frac{1}{2}$  is preferred over the observed Hessian  $\frac{1}{2}(y - \mu)^2 / \sigma^2$  because it is always positive and gives more stable tree splits.

### 4.3 Implementation Differences

- **XGBoost** (Chen & Guestrin, 2016): exact or approximate greedy split finding, L1/L2 regularization on leaf weights.
  - **LightGBM** (Ke et al., 2017): histogram-based splits, gradient-based one-side sampling (GOSS), exclusive feature bundling (EFB).
  - **CatBoost** (Prokhorenkova et al., 2018): ordered boosting to prevent target leakage, oblivious (symmetric) trees with balanced splits at each depth.
- Uncertainty:** Learned  $\sigma^2(\mathbf{x})$  via custom NLL objective, or post-hoc BINNED\_RESIDUAL\_STD.

## 5 Plain PyTorch Neural Network

A feedforward MLP:

$$f_{\theta}(\mathbf{x}) = W_L \cdot \sigma(W_{L-1} \cdots \sigma(W_1 \mathbf{x} + b_1) \cdots + b_{L-1}) + b_L,$$

where  $\sigma$  is an activation function (ReLU, GELU, etc.) and  $L$  is the number of layers.

**Uncertainty modes:**

- **CONSTANT:** No learned uncertainty.  $\hat{\sigma}^2$  is a global constant.
- **PROBABILISTIC:** Output layer produces  $(\mu, \log \sigma^2)$ ; trained with Gaussian NLL (§2.2).
- **MC\_DROPOUT:** Dropout active at test time; run  $M$  forward passes (§2.4).
- **BINNED\_RESIDUAL\_STD:** Post-hoc binned variance (§2.5).

## 6 Classifier-Regression Model

A two-stage model: (1) discretize the target into bins and train a classifier, (2) map classifier probabilities back to regression predictions.

### 6.1 Architecture Overview

### 6.2 Binning

Targets are binned using percentile-based boundaries:

$$q_j = \text{Percentile} \left( y_{\text{train}}, \frac{100 \cdot j}{K} \right), \quad j = 0, 1, \dots, K, \quad (7)$$

$$c_i = \max\{j : y_i \geq q_j\}, \quad c_i \in \{0, 1, \dots, K-1\}. \quad (8)$$

Duplicate boundaries (from repeated values) are collapsed, and  $K$  is reduced accordingly.

#### Implementation Note

The collapse is *silent*: on discrete or heavily tied targets the effective  $K$  can be smaller than requested, with no warning emitted. Callers should check the returned boundary count rather than assume the requested  $K$ .

### 6.3 Stage 1: Classification

Train a classifier  $f : \mathbb{R}^d \rightarrow \Delta^K$  (NN, XGBoost, LightGBM, or CatBoost) on  $(\mathbf{x}_i, c_i)$  pairs. After training, **freeze** the classifier.

### 6.4 Stage 2: Mapping

A *mapper* transforms class probabilities to regression predictions.

### 6.4.1 Linear Mapper

$\hat{y} = a \cdot \hat{\pi} + b$ , where  $\hat{\pi}$  is a scalar summary of the probability vector and  $(a, b)$  are fit by OLS.

**Uncertainty:** Constant residual variance  $\hat{\sigma}^2 = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2$ .

### 6.4.2 Lookup Mapper (LOOKUP\_MEDIAN / LOOKUP\_MEAN)

The most reliable mapper. Bins the probability space and computes per-bin statistics:

1. For each class  $k$ , collect  $\pi_k(\mathbf{x}_i)$  across training samples.
2. Bin the probability values into  $B$  bins (where  $B = \max(B_{\min}, \min(B_{\max}, N/100))$ ).
3. For each bin  $b$ :

$$\hat{y}_b = \text{median}(\{y_i : \pi_k(\mathbf{x}_i) \in b\}) \quad (\text{or mean}), \quad (9)$$

$$\hat{\sigma}_b^2 = \text{var}(\{y_i : \pi_k(\mathbf{x}_i) \in b\}) \quad (\text{if PROBABILISTIC}). \quad (10)$$

4. At prediction time: look up the bin for  $\pi_k(\mathbf{x})$ , return  $(\hat{y}_b, \hat{\sigma}_b^2)$ .

**Uncertainty:** Per-probability-bin variance — *input-dependent* through the probability: different  $\pi_k$  values land in different bins.

### 6.4.3 Spline Mapper

$\hat{y} = S(\pi_k; \text{degree} = k_s, \text{smoothing} = s)$  via `UnivariateSpline`. Known to be unstable.

**Uncertainty:** Constant residual variance (no per-input variance).

### 6.4.4 Neural Network Mapper

$(\hat{y}, \log \hat{\sigma}^2) = h_{\theta}(\pi)$  where  $h$  is a small MLP. Trained with NLL loss (probabilistic) or MSE (deterministic).

**Uncertainty:** Learned input-dependent  $\hat{\sigma}^2$  via NLL. Known training issues due to two-stage optimization.

## 7 Probabilistic Regression Model (ProbReg)

The central idea: discretize the prediction task into  $K$  classes via a learned classifier, attach a per-class regression head that outputs  $(\mu_k, \log \sigma_k^2)$ , and combine via the **law of total variance**.

### 7.1 Architecture Overview

ProbReg architecture. Solid arrows: data flow. Dashed: optional dynamic- $K$  path. Each regression head  $h_k$  receives only its own probability  $\pi_k$  (SEPARATE\_HEADS strategy).

1. **Classifier backbone**  $f_{\theta}: \mathbb{R}^d \rightarrow \mathbb{R}^K$ . Produces raw logits  $\ell_1, \dots, \ell_K$ . Softmax gives class probabilities:

$$\pi_k(\mathbf{x}) = \frac{\exp(\ell_k(\mathbf{x}))}{\sum_{j=1}^K \exp(\ell_j(\mathbf{x}))}.$$

2. **Regression module** (one of three strategies, §7.3).
3. **Aggregation** via law of total variance (§2.3).
4. **Optional: dynamic  $K$**  via a gating subnetwork (§9).

## 7.2 Class Masking for Variable $K$

When using dynamic  $K$  or a fixed  $K < K_{\max}$ , classes  $k > K$  are masked:

$$\tilde{\ell}_k = \begin{cases} \ell_k & \text{if } k \leq K, \\ -\infty & \text{if } k > K. \end{cases}$$

### Implementation Note

Masking shares the logits across  $K$ , but the percentile *bin boundaries* do not nest: they are recomputed independently for each candidate  $K$  (the  $K=2$  edges are not a subset of the  $K=3$  edges), so “class  $k$ ” denotes a different slice of the target at each  $K$ . This is harmless under likelihood-only training (`REGRESSION_ONLY`), where boundaries never enter the loss, but it makes the cross-entropy term unsound under dynamic  $K$  — see the caveat at Eq. (21).

## 7.3 Regression Strategies

Given class probabilities  $\pi_1, \dots, \pi_K$ , the regression module maps these to per-class outputs  $(\mu_k, \log \sigma_k^2)$  for  $k \in [K]$ .

### 7.3.1 Strategy 1: Separate Heads (`SEPARATE_HEADS`)

Each class  $k$  has an *independent* neural network head  $h_k$ :

$$h_k : \mathbb{R} \rightarrow \mathbb{R}^2, \quad h_k(\pi_k) = (\mu_k, \log \sigma_k^2).$$

The input to head  $k$  is the *scalar* probability  $\pi_k(\mathbf{x})$ . Heads do not share parameters.

### Implementation Note

This is the primary strategy used in benchmarks. The key architectural choice: each head only sees its own class probability, not the full probability vector. The classifier acts as a nonlinear feature extractor projecting onto the probability simplex; each head learns a scalar-to-scalar function.

### 7.3.2 Strategy 2: Single Head, $N$ Outputs (`SINGLE_HEAD_N_OUTPUTS`)

One shared neural network takes the full probability vector:

$$h : \mathbb{R}^K \rightarrow \mathbb{R}^{K \times 2}, \quad h(\boldsymbol{\pi}) = \{(\mu_k, \log \sigma_k^2)\}_{k=1}^K.$$

### 7.3.3 Strategy 3: Single Head, Final Output (`SINGLE_HEAD_FINAL_OUTPUT`)

One shared network maps probabilities to per-class representations:

$$h : \mathbb{R}^K \rightarrow \mathbb{R}^{K \times 2}, \quad \hat{y} = \sum_{k=1}^K \pi_k \cdot h_k(\boldsymbol{\pi}).$$

**Remark 7.1.** Strategies 2 and 3 allow parameter sharing across classes but lose the per-class specialization that makes `SEPARATE_HEADS` effective.

## 7.4 Law of Total Variance Aggregation

**Definition 7.1** (Law of Total Variance Decomposition).

$$\hat{y}(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{x}] = \sum_{k=1}^K \pi_k(\mathbf{x}) \cdot \mu_k(\mathbf{x}) \quad (11)$$

$$\hat{\sigma}^2(\mathbf{x}) = \text{Var}(Y \mid \mathbf{x}) = \underbrace{\sum_{k=1}^K \pi_k(\mathbf{x}) \cdot \sigma_k^2(\mathbf{x})}_{\text{expected aleatoric variance}} + \underbrace{\sum_{k=1}^K \pi_k(\mathbf{x}) \cdot (\mu_k(\mathbf{x}) - \hat{y}(\mathbf{x}))^2}_{\text{variance of means (epistemic-like)}} \quad (12)$$

### Implementation Note

In `pytorch_utils.apply_law_of_total_variance`: input shapes: probabilities  $(B, K)$ , per-head outputs  $(B, K, 2)$ . Output:  $(\hat{y}, \log \hat{\sigma}^2)$  with shape  $(B, 2)$ , where  $\log \hat{\sigma}^2 = \log(\max(\hat{\sigma}^2, 10^{-9}))$ .

## 7.5 Optimization Strategies

When `optimization_strategy = GRADIENT_STOP`:  $\tilde{\pi}_k = \text{stopgrad}(\pi_k)$ . The classifier is updated only by cross-entropy; regression heads only by NLL.

When `optimization_strategy = REGRESSION_ONLY`: the cross-entropy term is disabled entirely and the classifier is trained only through the combined predictive likelihood (the gradient of  $\mathcal{L}_{\text{regression}}$  flowing back through the class probabilities). No class-membership targets — and hence no bin boundaries — enter the loss. This is the *validated configuration for dynamic K*: it is the training mode of the two-stage recipe of record (frozen likelihood-only fit, then a post-hoc distilled selector), and it is structurally exempt from the class-identity conflict described at Eq. (21).

## 7.6 Monotonic Constraints

Separate mean and variance heads with monotonic linear on the mean:

$$\mu_k = g_\mu(\pi_k; \mathbf{w}_\mu) \quad (\text{monotonic}), \quad \log \sigma_k^2 = g_\sigma(\pi_k; \mathbf{w}_\sigma) \quad (\text{unconstrained}). \quad (13)$$

Monotonic linear:

$$f_{\text{mono}}(\mathbf{x}; \mathbf{w}, b) = \begin{cases} \text{softplus}(\mathbf{w})^\top \mathbf{x} + b & \text{positive monotonicity,} \\ -\text{softplus}(\mathbf{w})^\top \mathbf{x} + b & \text{negative monotonicity,} \end{cases}$$

where  $\text{softplus}(w) = \log(1 + e^w)$  ensures non-negative weights.

## 7.7 Boundary Constraints

`HARDSIGMOID` constrains predictions to  $[y_{\min}, y_{\max}]$ :

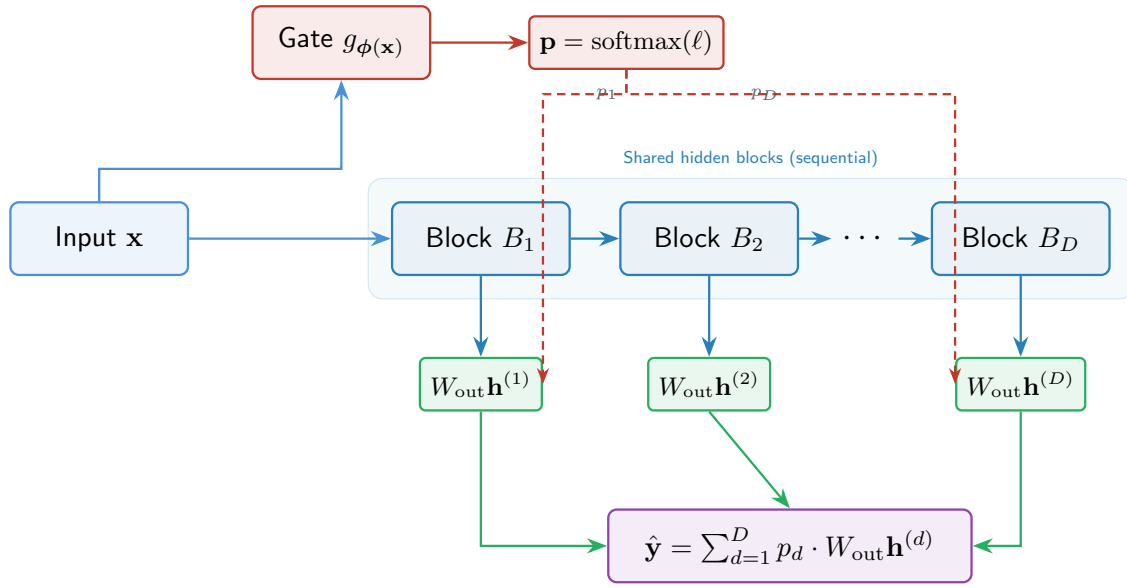
$$\tilde{\mu}_k = y_{\min} + (y_{\max} - y_{\min}) \cdot \text{hardsigmoid}(\mu_k).$$

**Uncertainty:** Law of total variance (§2.3) — the central novelty of ProbReg.

## 8 Flexible Neural Network (FlexibleNN)

A neural network with *learnable depth*: a gating subnetwork predicts how many hidden layers each input should use.

## 8.1 Architecture



Hard inference:  $d^* = \arg \max_d p_d$ , run only blocks  $B_1, \dots, B_{d^*}$

Figure 2: FlexibleNN architecture. Soft forward: all depth paths computed, weighted by  $p_d$ . Hard inference: only the argmax depth path is computed per sample.

- **Gate network**  $g_{\phi}: \mathbb{R}^d \rightarrow \mathbb{R}^{D_{\max}}$ .
- **Hidden blocks**  $B_1, \dots, B_{D_{\max}}$ : each is Linear  $\rightarrow$  BatchNorm  $\rightarrow$  Activation  $\rightarrow$  Dropout.
- **Output layer**  $W_{\text{out}}: \mathbb{R}^h \rightarrow \mathbb{R}^o$ , shared across all depths.

## 8.2 Soft Forward Pass

Depth probabilities:  $\mathbf{p} = \text{softmax}(g_{\phi(\mathbf{x})}) \in \Delta^{D_{\max}}$ .

For each depth  $d \in [D_{\max}]$ :  $\mathbf{h}^{(d)} = B_d \circ \dots \circ B_1(\mathbf{x})$ .

Aggregated output:

$$\hat{\mathbf{y}} = \sum_{d=1}^{D_{\max}} p_d(\mathbf{x}) \cdot W_{\text{out}} \mathbf{h}^{(d)}. \quad (14)$$

Training cost is  $O(D_{\max})$  per sample — all depth paths are computed.

## 8.3 Hard Forward Pass (Inference)

$d^*(\mathbf{x}) = \arg \max_d p_d(\mathbf{x})$ . Group samples by depth, run only needed blocks:

---

**Algorithm 1** Hard inference for FlexibleNN

---

```

1: for  $d = 1$  to  $D_{\max}$  do
2:    $\mathcal{I}_d \leftarrow \{i : d^*(\mathbf{x}_i) = d\}$ 
3:   if  $\mathcal{I}_d \neq \emptyset$  then
4:      $\hat{\mathbf{y}}_{\mathcal{I}_d} \leftarrow W_{\text{out}}(B_d \circ \dots \circ B_1(\mathbf{x}_{\mathcal{I}_d}))$ 
5:   end if
6: end for
```

---

Compute savings:  $(D_{\max} - \bar{d})/D_{\max}$ . Accuracy gap:  $\sim 0.02$  MSE.

## 8.4 Independent-Weights Variant

Each depth option has a *completely independent* network  $f_d$  with exactly  $d$  hidden layers (no shared blocks):

$$\hat{\mathbf{y}} = \sum_{d=1}^{D_{\max}} p_d(\mathbf{x}) \cdot f_d(\mathbf{x}).$$

This is a **mixture of experts**. More parameters ( $O(h^2 \cdot D_{\max}^2/2)$  vs  $O(h^2 \cdot D_{\max})$ ) but full per-depth specialization.

**Uncertainty:** Same as Plain NN (depends on `uncertainty_method`).

## Part III

# Dynamic Architecture Selection

The framework below is shared by ProbReg (selecting number of classes  $K$ ) and FlexibleNN (selecting depth  $D$ ). We use  $z$  as the generic discrete latent variable.

## 9 Selection Strategies

All strategies compute a probability vector  $\mathbf{p} \in \Delta^{|\mathcal{Z}|}$  over possible values of  $z$ , then aggregate predictions.

### 9.1 None Strategy (Fixed)

$p_k = \delta_{k, k_{\text{fixed}}}$ . No gating network.

### 9.2 Soft-Gating

Fully differentiable:

$$\mathbf{p} = \text{softmax}(g_{\phi(\mathbf{x})}), \quad \hat{\mathbf{y}} = \sum_k p_k \cdot \hat{\mathbf{y}}^{(k)}.$$

#### Implementation Note

SoftGating is the best-performing strategy for ProbReg dynamic- $K$ . It achieves MSE matching the best fixed- $K$  baseline with mean  $K \approx 2$ . Caveat: this ablation predates the dynamic- $K$  class-identity finding (see Eq. (21)); for deployment, dynamic  $K$  is validated only with likelihood-only training (REGRESSION\_ONLY).

### 9.3 Gumbel-Softmax

**Training:**  $p_k = \exp((\ell_k + G_k)/\tau) / \sum_j \exp((\ell_j + G_j)/\tau)$ ,  $G_k \sim \text{Gumbel}(0, 1)$ .

**Evaluation:**  $p_k = \exp(\ell_k/\tau) / \sum_j \exp(\ell_j/\tau)$  (deterministic).

As  $\tau \rightarrow 0$ , samples approach one-hot. Aggregation same as Soft-Gating.

**Remark 9.1.** Gumbel-Softmax for dynamic  $K$  has poor empirical performance — the stochastic noise corrupts KL gradients in the ELBO formulation.

### 9.4 Straight-Through Estimator (STE)

**Forward:**  $\mathbf{p}_{\text{hard}} = \text{one\_hot}(\arg \max_k \ell_k)$ .

**Backward:** Gradients flow through the soft distribution:  $\mathbf{p}_{\text{STE}} = \mathbf{p}_{\text{hard}} - \text{stopgrad}(\mathbf{p}_{\text{soft}}) + \mathbf{p}_{\text{soft}}$ .

**Remark 9.2.** The weighted-sum form is essential. Overwriting entries of  $\mathbf{p}_{\text{soft}}$  with hard one-hot values via indexed assignment severs the gradient path through the gating parameters and makes the strategy equivalent to a frozen argmax.

## 9.5 REINFORCE

Discrete sampling with policy gradient:

$$k^* \sim \text{Categorical}(\text{softmax}(g_{\phi(\mathbf{x})})), \quad \hat{\mathbf{y}} = \hat{\mathbf{y}}^{(k^*)}.$$

Policy gradient loss:

$$\mathcal{L}_{\text{REINFORCE}} = -\log \pi_{k^*}(\mathbf{x}) \cdot R, \quad R = -\mathcal{L}_{\text{task}}^{\text{detach}}. \quad (15)$$

**Remark 9.3.** REINFORCE has high variance. Needs baseline subtraction  $R \leftarrow R - \bar{R}$  (not yet implemented).

## 9.6 Strategy Summary

| Strategy       | Forward Pass                                      | Gradient                         |
|----------------|---------------------------------------------------|----------------------------------|
| None           | $p_k = \delta_{k, k_{\text{fixed}}}$              | N/A                              |
| Soft-Gating    | $\mathbf{p} = \text{softmax}(\ell)$               | Standard softmax gradient        |
| Gumbel-Softmax | $p_k \propto e^{(\ell_k + G_k)/\tau}$             | Reparameterization               |
| STE            | Forward:<br>one_hot(arg max); back-<br>ward: soft | Straight-through                 |
| REINFORCE      | $k^* \sim \text{Cat}(\text{softmax}(\ell))$       | $-R \cdot \nabla \log \pi_{k^*}$ |

## 10 Regularization for Dynamic Selection

Without regularization, the gating network drifts to  $z = z_{\text{max}}$ .

### 10.1 Penalty

Penalize expected complexity:

$$\mathcal{L}_{\text{penalty}} = \lambda \cdot \sum_k k \cdot p_k(\mathbf{x}). \quad (16)$$

Used as K\_PENALTY (ProbReg) and DEPTH\_PENALTY (FlexibleNN).

### 10.2 ELBO with KL Divergence

Treat  $z$  as a discrete latent variable with prior  $p(z)$  favoring simplicity.

**Prior:**

$$p(z = k) \propto w_k, \quad (w_1, \dots, w_{|\mathcal{Z}|}) = \text{linspace}(3.0, 1.0, |\mathcal{Z}|).$$

The `linspace(3, 1)` parameterization ensures constant steepness regardless of  $|\mathcal{Z}|$ .

**Posterior:**  $q(z = k | \mathbf{x}) = p_k(\mathbf{x}) = \text{softmax}(g_{\phi(\mathbf{x})})_k$ .

**ELBO:**

$$\mathcal{L}_{\text{ELBO}} = \mathcal{L}_{\text{task}} + \lambda_{\text{KL}} \cdot \text{KL}(q(z | \mathbf{x}) \| p(z)), \quad (17)$$

$$\text{KL}(q \| p) = \sum_k q(z = k | \mathbf{x}) [\log q(z = k | \mathbf{x}) - \log p(z = k)]. \quad (18)$$

### 10.3 Unified View: ELBO as Occam's Razor

This is the same principle as in variational autoencoders (Kingma & Welling, 2014), but applied to a *discrete architectural choice* rather than a continuous latent code.

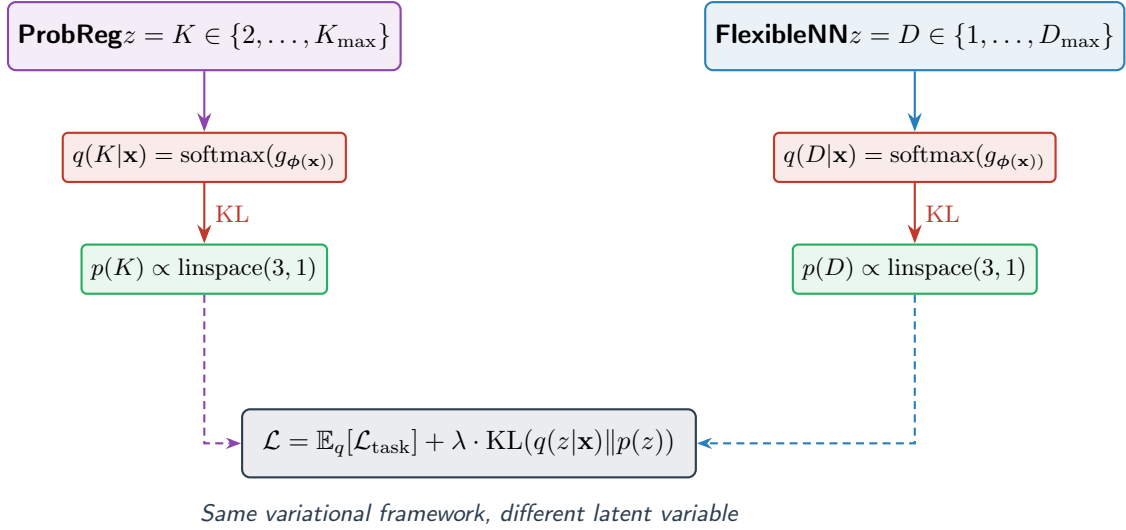


Figure 3: Unified ELBO framework. ProbReg selects  $K$ ; FlexibleNN selects  $D$ . Both use the same KL penalty against a linspace prior.

#### 10.4 Direct Regression Channel (ProbReg only)

When dynamic  $K$  is enabled, index 0 of the gating output corresponds to a *direct regression* bypass:  $\hat{\mathbf{y}}^{(0)} = f_{\text{direct}}(\mathbf{x})$ , allowing fallback to standard regression when the classification bottleneck adds no value.

## Part IV

# Loss Functions and Training

## 11 Loss Functions

### 11.1 Gaussian Negative Log-Likelihood (NLL)

$$\mathcal{L}_{\text{NLL}}(\hat{y}, \hat{\sigma}^2, y) = \frac{1}{2} \left( \log(2\pi) + \log \hat{\sigma}^2 + \frac{(y - \hat{y})^2}{\hat{\sigma}^2} \right). \quad (19)$$

### 11.2 $\beta$ -NLL Loss (Seitzer et al., 2022)

$$\mathcal{L}_{\beta\text{-NLL}} = (\hat{\sigma}_{\text{detach}}^2)^\beta \cdot \frac{1}{2} \left( \log \hat{\sigma}^2 + \frac{(y - \hat{y})^2}{\hat{\sigma}^2} \right). \quad (20)$$

$\beta = 0$ : standard NLL.  $\beta = 0.5$ : recommended default.  $\beta = 1$ : can over-correct.

**Intuition.** Large  $\hat{\sigma}^2$  amplifies the loss via  $(\hat{\sigma}^2)^\beta$ , discouraging variance inflation.

### 11.3 Masked Cross-Entropy Loss

$$\mathcal{L}_{\text{CE}} = -\log \frac{\exp(\tilde{\ell}_{c^*})}{\sum_j \exp(\tilde{\ell}_j)},$$

where  $c^*$  is the true bin index and  $\tilde{\ell}$  are masked logits. Under dynamic  $K$ ,  $c^*$  is computed against the boundary set of the  $K$  that sample selected on that forward pass; since boundary sets for different  $K$  are not refinements of one another,  $c^*$  is not a stable class identity across  $K$  (see below).

### 11.4 Combined Training Loss (ProbReg)

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{regression}} + \lambda_{\text{CE}} \cdot \mathcal{L}_{\text{CE}} + \lambda_{\text{reg}} \cdot \mathcal{L}_{\text{regularization}} + \lambda_{\text{boundary}} \cdot \mathcal{L}_{\text{boundary}} + \mathcal{L}_{\text{selection}}. \quad (21)$$

Currently  $\lambda_{\text{CE}} = 1$  (implicit, not tuned).

#### Implementation Note

**Caveat (dynamic  $K$ , verified 2026-07).** With dynamic  $K$  the  $\mathcal{L}_{\text{CE}}$  term is structurally unsound: each sample’s target  $c^*$  is built from the percentile boundaries of that sample’s selected  $K$ , and those boundary sets do not nest across  $K$ . Logit 0 is therefore supervised as “bottom  $1/K$  of  $y$ ” for whatever  $K$  each sample selects — the same component asked to mean different things at different resolutions, with nothing in the input distinguishing the regime. The validated configuration for dynamic  $K$  is **REGRESSION\_ONLY** (cross-entropy disabled; see *Optimization Strategies*), under which class identity is carried by the shared masked-prefix logits alone and remains stable across  $K$ . Cross-entropy-active dynamic  $K$  should be treated as unsupported pending a  $K$ -stable class semantics (e.g. boundaries anchored to a single  $K_{\text{max}}$  grid, or refinement binning).

## 11.5 Boundary Regularization Loss

$$\mathcal{L}_{\text{boundary}} = \frac{1}{N} \sum_{i=1}^N [\max(0, y_{\min} - \hat{y}_i)^2 + \max(0, \hat{y}_i - y_{\max})^2].$$

## 11.6 Middle-Class Penalty

When  $K$  is odd, penalize the middle class  $k_{\text{mid}} = \lceil K/2 \rceil$ :

$$\mathcal{L}_{\text{mid}} = -\log \mathcal{N}(\mu_{k_{\text{mid}}}; \bar{y}_{\text{mid}}, \sigma_{k_{\text{mid}}}^2).$$

## 11.7 Tree-Model NLL Objective

See §4.2 (in the GBT section) for the custom gradients and Hessians used by XGBoost, LightGBM, and CatBoost.

# 12 Learned Regularization

Instead of fixing  $\lambda_1, \lambda_2$ , they are learned via separate optimizer steps. Let  $d$  be the number of model parameters.

### 12.1 L1 Only (Laplace Prior)

$$\mathcal{L}_{\text{reg}} = -d \cdot \log \frac{\lambda_1}{2} + \lambda_1 \|\mathbf{w}\|_1, \quad \lambda_1 = e^{\ell_1}.$$

### 12.2 L2 Only (Gaussian Prior)

$$\mathcal{L}_{\text{reg}} = -\frac{d}{2} \log \frac{\lambda_2}{\pi} + \lambda_2 \|\mathbf{w}\|_2^2.$$

### 12.3 L1 + L2 (Horseshoe-like Prior)

$$\mathcal{L}_{\text{reg}} = d \cdot \log Z + \lambda_1 \|\mathbf{w}\|_1 + \lambda_2 \|\mathbf{w}\|_2^2, \quad \log Z = \frac{1}{2} \log \frac{\pi}{\lambda_2} + \frac{\lambda_1^2}{4\lambda_2} + \log \text{erfc}\left(\frac{\lambda_1}{2\sqrt{\lambda_2}}\right).$$

# 13 Target Transforms

## 13.1 Symmetric Log Transform (Symlog)

For targets spanning multiple orders of magnitude:

$$\text{symlog}(x) = \text{sign}(x) \cdot \log(1 + |x|), \tag{22}$$

$$\text{symexp}(y) = \text{sign}(y) \cdot (\exp(|y|) - 1). \tag{23}$$

$\text{symexp}$  is the exact inverse. Compresses large values ( $\text{symlog}(1000) \approx 6.9$ ) while preserving sign.

## 13.2 Uncertainty Conversion: Monte Carlo

If the model predicts  $(\mu_s, \sigma_s^2)$  in symlog space:

1. Sample  $\epsilon_1, \dots, \epsilon_M \sim \mathcal{N}(\mu_s, \sigma_s^2)$ ,  $M = 200$ .
2. Apply:  $z_m = \text{symexp}(\epsilon_m)$ .
3. Return:  $\hat{\mu} = \frac{1}{M} \sum z_m$ ,  $\hat{\sigma} = \sqrt{\frac{1}{M} \sum (z_m - \hat{\mu})^2}$ .

Exact up to MC variance; handles nonlinearity of  $\text{symexp}$  near zero.

## 14 Training Loop

All neural models follow this loop (`base_pytorch.py`):

---

**Algorithm 2** Training loop for PyTorch models

---

**Require:** Model  $f_{\theta}$ , data  $(\mathbf{X}, \mathbf{y})$ , learning rate  $\eta$ , patience  $P$

```

1: Split into train/validation
2: Initialize Adam optimizer
3:  $\text{best\_loss} \leftarrow \infty$ ,  $\text{wait} \leftarrow 0$ 
4: for epoch = 1 to  $\text{max\_epochs}$  do
5:   for each mini-batch  $(\mathbf{X}_b, \mathbf{y}_b)$  do
6:      $\hat{\mathbf{y}}_b \leftarrow f_{\theta}(\mathbf{x}_b)$ 
7:      $\mathcal{L} \leftarrow \text{criterion}(\hat{\mathbf{y}}_b, \mathbf{y}_b) + \mathcal{L}_{\text{reg}}(\theta)$ 
8:     Backprop and update  $\theta$  via Adam
9:     if learned regularization then update  $\lambda$  params separately
10:    end if
11:  end for
12:   $\mathcal{L}_{\text{val}} \leftarrow$  evaluate on validation set
13:  if  $\mathcal{L}_{\text{val}} < \text{best\_loss}$  then
14:    Save checkpoint,  $\text{wait} \leftarrow 0$ 
15:  else
16:     $\text{wait} \leftarrow \text{wait} + 1$ 
17:    if  $\text{wait} \geq P$  then break
18:    end if
19:  end if
20: end for
21: Load best checkpoint

```

---

## Part V

### Baseline Models

#### 15 NGBoost (Duan et al., 2020)

Natural gradient boosting. For  $\mathcal{N}(\mu, \sigma^2)$ , update at each step uses the *natural gradient*:

$$\tilde{\nabla}_{\boldsymbol{\theta} \mathcal{L} = \mathcal{I}(\boldsymbol{\theta})^{-1} \nabla_{\boldsymbol{\theta} \mathcal{L}}, \quad \mathcal{I} = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 1/(2\sigma^4) \end{pmatrix}.$$

This accounts for the geometry of the parameter space, preventing over-aggressive  $\sigma$  updates.

**Uncertainty:** Learned  $\hat{\sigma}^2(\mathbf{x})$ , input-dependent via tree structure. Primary tree-based competitor to ProbReg.

#### 16 Mixture Density Network — MDN (Bishop, 1994)

A neural network outputting  $M$ -component Gaussian mixture parameters:

$$p(y | \mathbf{x}) = \sum_{m=1}^M w_m(\mathbf{x}) \cdot \mathcal{N}(y; \mu_m(\mathbf{x}), \sigma_m^2(\mathbf{x})), \quad (24)$$

with  $w_m = \text{softmax}(\ell^w)_m$ ,  $\sigma_m = \exp(\ell_m^\sigma/2)$ .

**Loss:**  $\mathcal{L}_{\text{MDN}} = -\frac{1}{N} \sum_i \log(\sum_m w_m(\mathbf{x}_i) \cdot \phi(y_i; \mu_m, \sigma_m^2))$ .

**Uncertainty:** Full mixture variance via law of total variance:  $\hat{\sigma}^2 = \sum_m w_m \sigma_m^2 + \sum_m w_m (\mu_m - \hat{\mu})^2$ .

**vs ProbReg:** MDN has no structural constraint on components; ProbReg imposes a classification bottleneck that regularizes the mixture.

#### 17 Deep Ensembles (Lakshminarayanan et al., 2017)

Train  $M$  independently initialized networks, each outputting  $(\mu_m, \sigma_m^2)$ :

$$\hat{\mu} = \frac{1}{M} \sum_{m=1}^M \mu_m, \quad \hat{\sigma}^2 = \frac{1}{M} \sum_{m=1}^M \sigma_m^2 + \frac{1}{M} \sum_{m=1}^M (\mu_m - \hat{\mu})^2. \quad (25)$$

Same law-of-total-variance decomposition as ProbReg with  $\pi_m = 1/M$ .

**vs ProbReg:**  $M$  full-size networks ( $O(M)$  forward passes) vs lightweight per-class heads behind a shared classifier ( $O(1)$  forward passes).

#### 18 Quantile Regression NN (Koenker, 2005)

Trained with pinball loss at multiple quantiles  $\tau_1 < \dots < \tau_Q$ :

$$\mathcal{L}_{\text{QR}} = \frac{1}{NQ} \sum_{i,j} L_{\tau_j}(y_i, \hat{q}_{\tau_j}(\mathbf{x}_i)), \quad L_{\tau}(y, \hat{q}) = \begin{cases} \tau(y - \hat{q}) & y \geq \hat{q}, \\ (1 - \tau)(\hat{q} - y) & y < \hat{q}. \end{cases}$$

Distribution-free. Intervals:  $[\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}]$ .

## 19 Conformalized Quantile Regression (Romano et al., 2019)

CQR calibrates the intervals produced by a quantile regressor (§18) via split conformal prediction (§25) so that coverage is distribution-free and finite-sample-valid.

Fit  $(\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2})$  on the training fold. On a held-out calibration set  $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$  compute the conformity scores

$$R_i = \max(\hat{q}_{\alpha/2}(\mathbf{x}_i) - y_i, y_i - \hat{q}_{1-\alpha/2}(\mathbf{x}_i)), \quad (26)$$

which measure how far each target falls outside the predicted interval (negative when inside). Take the finite-sample-corrected quantile

$$\hat{Q}_\alpha = \text{Quantile}\left(\{R_i\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right).$$

**Calibrated interval:**

$$C(\mathbf{x}) = [\hat{q}_{\alpha/2}(\mathbf{x}) - \hat{Q}_\alpha, \hat{q}_{1-\alpha/2}(\mathbf{x}) + \hat{Q}_\alpha].$$

Coverage  $\Pr(y \in C(\mathbf{x})) \geq 1 - \alpha$  holds under exchangeability. Unlike marginal split conformal (§25), the CQR interval width varies with  $\mathbf{x}$  because the base quantile regressor is heteroscedastic.

## 20 Gaussian Process

$$\hat{\mu}(\mathbf{x}^*) = \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y}, \quad (27)$$

$$\hat{\sigma}^2(\mathbf{x}^*) = k(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_*. \quad (28)$$

Training  $O(N^3)$ ; limited to  $N \leq 5000$ .

## 21 Constant-Variance NN

$f_{\theta(\mathbf{x})=(\mu(\mathbf{x}), \log \sigma^2)}$  where  $\log \sigma^2$  is a single global parameter.

### Implementation Note

Implemented as ProbReg with  $K = 1$ , `uncertainty_method = PROBABILISTIC`.

## 22 FT-Transformer (Gorishniy et al., 2021)

Tabular transformer: tokenizes each feature into an embedding and applies multi-head self-attention.

**Feature tokenization:** For feature  $j$ :  $\mathbf{e}_j = \mathbf{w}_j \cdot x_j + \mathbf{b}_j \in \mathbb{R}^{d_{\text{emb}}}$ . Prepend [CLS] token  $\mathbf{e}_0$ .

**Transformer blocks** ( $L$  layers):

$$\tilde{\mathbf{E}}^{(l)} = \mathbf{E}^{(l-1)} + \text{MultiHeadAttn}(\text{LN}(\mathbf{E}^{(l-1)})), \quad (29)$$

$$\mathbf{E}^{(l)} = \tilde{\mathbf{E}}^{(l)} + \text{FFN}(\text{LN}(\tilde{\mathbf{E}}^{(l)})). \quad (30)$$

**Prediction:**  $\hat{y} = \mathbf{w}_{\text{out}}^\top \mathbf{e}_0^{(L)} + b_{\text{out}}$ .

**Uncertainty:** No native UQ. Can combine with Deep Ensembles, MC Dropout, or conformal prediction.

**Proposed extension (Paper C):** Apply FlexibleNN depth gating to FT-Transformer blocks — variable self-attention depth per input with ELBO regularization.

## 23 TabPFN (Hollmann et al., 2024)

Pre-trained transformer performing in-context learning:

$$\hat{y}(\mathbf{x}^*) = \text{TabPFN}(\mathbf{x}^*, \mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}).$$

No task-specific training. Pre-trained on millions of synthetic datasets.

**Uncertainty:** Outputs a predictive distribution (quantile predictions for intervals).

**Limits:**  $N \leq 10,000$ ,  $d \leq 500$ .

## 24 Laplace-Approximation BNN (Daxberger et al., 2021)

*Not yet implemented — included for comparison completeness.* Post-hoc Bayesian: train MAP estimate  $\boldsymbol{\theta}_{\text{MAP}}$ , approximate posterior as Gaussian:  $p(\boldsymbol{\theta} \mid \mathcal{D}) \approx \mathcal{N}(\boldsymbol{\theta}_{\text{MAP}}, (\nabla^2 \mathcal{L})^{-1})$ .

**Predictive variance** via linearization:

$$\hat{\sigma}^2(\mathbf{x}^*) = \sigma_n^2 + \mathbf{J}(\mathbf{x}^*)^\top (\nabla^2 \mathcal{L})^{-1} \mathbf{J}(\mathbf{x}^*), \quad (31)$$

where  $\mathbf{J}(\mathbf{x}^*) = \nabla_{\boldsymbol{\theta}} f(\mathbf{x}^*; \boldsymbol{\theta}_{\text{MAP}})$ .

Available via `laplace-torch` (~50 lines wrapper).

## 25 Conformal Prediction

Distribution-free intervals that wrap *any* fitted model. Given a calibration set  $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$  and miscoverage  $\alpha$ .

### 25.1 Split Conformal (Marginal)

1. Nonconformity scores:  $R_i = |y_i - \hat{f}(\mathbf{x}_i)|$ .
2. Finite-sample-corrected quantile:

$$\hat{q}_\alpha = \text{Quantile} \left( \{R_i\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right). \quad (32)$$

3. Prediction interval:  $C(\mathbf{x}) = [\hat{f}(\mathbf{x}) - \hat{q}_\alpha, \hat{f}(\mathbf{x}) + \hat{q}_\alpha]$ .

**Definition 25.1** (Coverage Guarantee). Under exchangeability:  $\Pr(y \in C(\mathbf{x})) \geq 1 - \alpha$ .

Interval width is a constant  $2\hat{q}_\alpha$  — over-covers in low-noise regions and under-covers in high-noise regions on a per-point basis.

### 25.2 Locally-Adaptive Conformal (Lei & Wasserman, 2014)

Given a base model that supplies both a point prediction  $\hat{f}(\mathbf{x})$  and an uncertainty estimate  $\hat{\sigma}(\mathbf{x})$ , normalize the nonconformity scores by the predicted standard deviation:

$$\tilde{R}_i = \frac{|y_i - \hat{f}(\mathbf{x}_i)|}{\hat{\sigma}(\mathbf{x}_i)}, \quad \hat{Q}_\alpha = \text{Quantile} \left( \{\tilde{R}_i\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right). \quad (33)$$

**Adaptive interval:**

$$C(\mathbf{x}) = [\hat{f}(\mathbf{x}) - \hat{Q}_\alpha \cdot \hat{\sigma}(\mathbf{x}), \hat{f}(\mathbf{x}) + \hat{Q}_\alpha \cdot \hat{\sigma}(\mathbf{x})].$$

The same coverage guarantee  $\Pr(y \in C(\mathbf{x})) \geq 1 - \alpha$  holds, but intervals now contract where the base model is confident and widen where it is uncertain. This gives *conditional* coverage if  $\hat{\sigma}(\mathbf{x})$  tracks the true residual scale.

## Part VI

### Evaluation

#### 26 Evaluation Metrics

##### 26.1 Point-Prediction Metrics

$$\text{MAE} = \frac{1}{N} \sum_i |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2}, \quad (34)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad \text{MAPE} = \frac{100}{N} \sum_i \frac{|y_i - \hat{y}_i|}{|y_i|}. \quad (35)$$

##### 26.2 Proper Scoring Rules

**Negative Log-Likelihood:**

$$\text{NLL} = \frac{1}{N} \sum_i \frac{1}{2} \left( \log(2\pi) + \log \hat{\sigma}_i^2 + \frac{(y_i - \hat{y}_i)^2}{\hat{\sigma}_i^2} \right).$$

**CRPS.** For  $\mathcal{N}(\mu, \sigma^2)$ :

$$\text{CRPS} = \sigma \left[ \frac{y - \mu}{\sigma} \left( 2\Phi\left(\frac{y - \mu}{\sigma}\right) - 1 \right) + 2\phi\left(\frac{y - \mu}{\sigma}\right) - \frac{1}{\sqrt{\pi}} \right]. \quad (36)$$

Strictly proper. Generalizes to non-Gaussian predictive distributions via numerical integration.

**Pinball (quantile) loss.**  $L_\tau(y, \hat{q}_\tau) = \max(\tau(y - \hat{q}_\tau), (1 - \tau)(\hat{q}_\tau - y))$ ; at  $\tau = 0.5$  equals MAE.

##### 26.3 Interval Metrics

**PICP.**  $\text{PICP}_\alpha = \frac{1}{N} \sum_i \mathbf{1}[y_i \in [\hat{y}_i - z_{\alpha/2} \hat{\sigma}_i, \hat{y}_i + z_{\alpha/2} \hat{\sigma}_i]]$ .

**MPIW.**  $\text{MPIW}_\alpha = \frac{1}{N} \sum_i (U_i - L_i)$ .

**Winkler score.** For interval  $[L, U]$  at level  $(1 - \alpha)$ :

$$W_\alpha = (U - L) + \frac{2}{\alpha} (L - y) \mathbf{1}[y < L] + \frac{2}{\alpha} (y - U) \mathbf{1}[y > U]. \quad (37)$$

Strictly proper scoring rule for intervals.

##### 26.4 Calibration via PIT (Kuleshov et al., 2018)

Probability Integral Transform:  $U_i = F(y_i \mid \mathbf{x}_i)$ . If the predictive distribution is calibrated,  $U \sim \text{Uniform}(0, 1)$ .

**Calibration curve:** for target quantiles  $p_0 < \dots < p_B \in [0, 1]$ ,  $\hat{p}_j = \frac{1}{N} \sum_i \mathbf{1}[U_i \leq p_j]$ .

**Regression ECE:**  $\text{ECE} = \frac{1}{B+1} \sum_j |p_j - \hat{p}_j|$ .

**Miscalibration area:**  $\int_0^1 |\hat{p}(t) - t| dt$  (trapezoidal). More robust than bin-averaged ECE.

**Sharpness:**  $\frac{1}{N} \sum_i \hat{\sigma}_i$ . Always reported jointly with calibration — a calibrated model with wide intervals is useless.

## 26.5 Post-hoc Recalibration

Isotonic regression from predicted CDF values to empirical CDF on a held-out set (Kuleshov et al., 2018). Let  $\{U_i^{\text{cal}}\}$  be PIT values on the calibration set. Fit a monotone  $\hat{R} : [0, 1] \rightarrow [0, 1]$  with

$$\hat{R}(U_{(i)}^{\text{cal}}) \approx i/n_{\text{cal}}.$$

At test time apply  $U_i^{\text{recal}} = \hat{R}(U_i)$ . Report metrics as “+recal” variants to separate raw calibration quality from post-hoc-fixable calibration quality.

## 26.6 PIT Uniformity Tests

Sharper than visual inspection of histograms. Let  $\{U_i\}$  be the PIT values.

**Kolmogorov-Smirnov:**  $\text{KS} = \sup_t |\hat{F}_N(t) - t|$  where  $\hat{F}_N$  is the empirical CDF of  $\{U_i\}$ .

**Cramér-von Mises:**  $\text{CvM} = \frac{1}{12N} + \sum_{i=1}^N \left( \frac{2i-1}{2N} - U_{(i)} \right)^2$ . More sensitive to global deviations.

**Anderson-Darling:** weights deviations by  $1/[t(1-t)]$ , emphasizing the tails of the distribution. Implemented by transforming  $Z_i = \Phi^{-1}(U_i)$  and applying the standard normal AD test, since SciPy does not expose AD directly for the uniform distribution.

## 27 Photometric Redshift Metrics

**Normalized residual:**  $\Delta z_i = (z_{\text{phot},i} - z_{\text{spec},i}) / (1 + z_{\text{spec},i})$ .

**Bias:**  $\langle \Delta z \rangle$ . **Scatter:**  $\sigma_{\text{MAD}} = 1.4826 \cdot \text{median}(|\Delta z_i - \text{median}(\Delta z)|)$ . **Outlier fraction:**  $\eta = \frac{1}{N} \sum_i \mathbf{1}[|\Delta z_i| > 0.15]$ .

**PIT uniformity:** KS, CvM, and AD tests on  $\{F(z_{\text{spec},i} | \mathbf{x}_i)\}$  vs Uniform(0, 1) (§26.4). CvM and AD are more sensitive to tail deviations, which drive outlier rates.

**CDE loss (Izbicki & Lee, 2017):**

$$\text{CDE\_loss} = -\frac{1}{N} \sum_i \log \hat{p}(z_{\text{spec},i} | \mathbf{x}_i), \quad (38)$$

the negative log-likelihood of the true redshift under the model’s predicted conditional density. Standard metric for full-PDF quality beyond point estimates.